

# Financial Correlation Networks: A Random Matrix Disparity Filter Approach with Ensemble Robustness Validation

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## Abstract

This paper presents a framework for extracting statistically significant economic interactions from multivariate financial return series and tracking their temporal evolution. Utilizing the full S&P 500 universe of  $N = 500$  constituents spanning a 10-year period ( $T = 2766$  observation days), we deploy Random Matrix Theory (RMT) to separate true economic signals from sampling noise. To capture local interactions without global bias, we implement the multiscale Disparity Filter ( $\alpha = 0.05$ ). The resulting backbone is a sparse network with 48 active nodes and 59 edges (network density  $\rho = 0.0523$ ). This backbone shows strong corporate clustering, yielding a modularity score of  $Q_{\text{mod}} = 0.5454$ . Laplacian spectral analysis, however, reveals that the network is globally fragmented (zero eigenvalue multiplicity = 8). Nevertheless, the algebraic connectivity of the Giant Connected Component (GCC) remains positive ( $\mu_2^{\text{GCC}} = 0.0297$ ), confirming that while the broader network is fragmented, the core 4.00% cluster maintains internal structural cohesion. Additionally, we implement overlapping rolling windows ( $T_w = 252$  trading days, step size = 21 days) to track the dynamic evolution of node centrality and community structure across varying macroeconomic regimes. These results highlight the limitations of local filtering in large financial datasets and provide a dynamic framework for analyzing systemic risk.

## 1 Introduction

Quantifying asset return dependencies is essential for portfolio optimization and monitoring systemic risk. Typically, these interactions are mapped using the Pearson correlation matrix. However, when the number of assets  $N$  is comparable to the number of time steps  $T$ , the sample correlation matrix gets distorted by sampling noise. This compromises the effectiveness of models such as Markowitz portfolio optimization, which conflate sampling noise with genuine economic signals.

To extract the underlying network structure, we employ two steps: spectral denoising and topological filtering. We combine Random Matrix Theory (RMT) with the Disparity Filter. We compare the resulting backbone to standard methods like global thresholding and the Minimum Spanning Tree (MST) to evaluate their efficacy in preserving topological information, clustering sectors, and handling shocks.

## 2 Theoretical Framework

### 2.1 Random Matrix Theory and the Marčenko-Pastur Law

Let  $\mathbf{R}$  be a  $T \times N$  matrix representing the return histories of  $N$  assets over  $T$  discrete observation intervals. Let  $P_i(t)$  denote the adjusted closing price of asset  $i \in \{1, \dots, N\}$  at time step

$t \in \{1, \dots, T\}$ . The continuously compounded log return series  $r_i(t)$  is given by:

$$r_i(t) = \ln P_i(t) - \ln P_i(t-1) \quad (1)$$

To ensure variance homogeneity across different asset classes, each return stream is standardized to zero mean and unit variance:

$$\tilde{r}_i(t) = \frac{r_i(t) - \langle r_i \rangle}{\sigma_i} \quad (2)$$

where  $\langle \cdot \rangle$  and  $\sigma_i$  denote the temporal sample mean and sample standard deviation, respectively, forcing  $\mathbb{E}[\tilde{r}_i] = 0$  and  $\mathbb{E}[\tilde{r}_i^2] = 1$ . The empirical sample correlation matrix  $\mathbf{C}$  is computed via the inner product:

$$\mathbf{C} = \frac{1}{T} \tilde{\mathbf{R}}^T \tilde{\mathbf{R}} \quad (3)$$

where  $\tilde{\mathbf{R}}$  is the  $T \times N$  standardized log-return matrix. The aspect ratio that controls the spectral limits is  $Q = T/N \geq 1$ .

If the variables are completely uncorrelated, the eigenvalue distribution of  $\mathbf{C}$  converges to the Marčenko-Pastur (MP) law as  $N, T \rightarrow \infty$ :

$$P_{\text{MP}}(\lambda) = \frac{Q}{2\pi\sigma^2} \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{\lambda} \quad (4)$$

where  $\lambda \in [\lambda_-, \lambda_+]$ . The theoretical bounds for the noise bulk are:

$$\lambda_{\pm} = \sigma^2 \left( 1 \pm \sqrt{\frac{1}{Q}} \right)^2 \quad (5)$$

where  $\sigma^2$  is the residual variance after removing the main collective modes. Eigenvalues inside  $[\lambda_-, \lambda_+]$  are just noise, while those above  $\lambda_+$  represent real macroeconomic factors.

To prevent the dominant market eigenvalue ( $\lambda_1$ ) from skewing the noise bulk, we use the decontamination method from Laloux et al. We isolate the market mode and adjust the residual variance to:

$$\sigma_{\text{adj}}^2 = \frac{N - \lambda_1}{N} \quad (6)$$

The random noise bulk bounds are subsequently recomputed using  $\sigma_{\text{adj}}^2$ , providing a threshold for separating noise from sector specific interactions.

## 2.2 The Inverse Participation Ratio (IPR)

To quantify the spatial localization and track the structural distribution of each eigenvector  $\mathbf{v}^{(k)}$ , we evaluate the Inverse Participation Ratio (IPR). For a given eigenvector  $k$ , the metric is defined as:

$$I_k = \sum_{i=1}^N [v_k(i)]^4 \quad (7)$$

where  $v_k(i)$  represents the  $i$ th component of the normalized  $k$ th eigenvector ( $\sum_{i=1}^N [v_k(i)]^2 = 1$ ). The theoretical limits of the IPR are bounded as:

$$\frac{1}{N} \leq I_k \leq 1 \quad (8)$$

**Proof of Bounds:** The lower bound is evaluated via the Cauchy Schwarz inequality applied to the vector of squared components  $\mathbf{x} = [v_k(1)^2, \dots, v_k(N)^2]^T$  and a uniform vector  $\mathbf{1} = [1, \dots, 1]^T$ :

$$\left( \sum_{i=1}^N v_k(i)^2 \cdot 1 \right)^2 \leq \left( \sum_{i=1}^N [v_k(i)]^4 \right) \left( \sum_{i=1}^N 1^2 \right) \implies (1)^2 \leq I_k \cdot N \implies I_k \geq \frac{1}{N} \quad (9)$$

This lower bound is uniquely satisfied when the eigenmode is completely delocalized, i.e., every corporate constituent contributes equally with  $v_k(i) = 1/\sqrt{N}$ . The upper bound  $I_k = 1$  occurs when the eigenmode is perfectly localized on a single asset ( $v_k(i) = 1$  for one node, and 0 elsewhere). Analyzing the IPR profile allows us to separate broad market trends from localized corporate noise.

### 2.3 The MultiScale Disparity Filter

Standard filtering uses a single global threshold to cut weak edges. This hurts small or localized clusters, often isolating specific sectors. To preserve multi scale interactions without introducing global bias, we implement the Disparity Filter (Serrano et al., 2009).

Consider a fully connected, weighted graph where the edge weights are mapped from the empirical correlations:  $W_{ij} = |C_{ij}|$  for  $i \neq j$ , and  $W_{ii} = 0$ . For each individual node  $i$ , its local interaction strength is defined as  $s_i = \sum_j W_{ij}$ , and its localized degree is  $k_i$ . Under the null hypothesis that the total strength  $s_i$  is distributed uniformly across its incident edges, we compute the probability that a specific weight configuration occurs by chance. The exact survival function computing the probability that an individual edge weight fractional contribution  $p_{ij} = W_{ij}/s_i$  occurs by pure chance is given by:

$$\alpha_{ij}^{(i)} = \left(1 - \frac{W_{ij}}{s_i}\right)^{k_i-1} \quad (10)$$

Because the graph is undirected, every edge is evaluated from the perspective of both terminal vertices. The final localized "p" value for an edge connecting node  $i$  and node  $j$  is defined as:

$$p_{ij} = \min\left(\alpha_{ij}^{(i)}, \alpha_{ji}^{(j)}\right) \quad (11)$$

An edge is structurally preserved in the filtered backbone if and only if it satisfies the localized significance condition:

$$p_{ij} < \alpha \quad (12)$$

where  $\alpha$  represents the significance threshold, fixed in this study at  $\alpha = 0.05$ .

### 2.4 Spectral Graph Theory and Complex Network Topologies

To evaluate the structural organization of the filtered networks, we introduce formal community detection and graph spectral operators. Modularity ( $Q_{\text{mod}}$ ) measures the structural strength of a network's division into localized communities or modules:

$$Q_{\text{mod}} = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m}\right) \delta(c_i, c_j) \quad (13)$$

where  $A_{ij}$  is the unweighted adjacency matrix ( $A_{ij} = 1$  if an edge exists, 0 otherwise),  $m = \frac{1}{2} \sum_i k_i$  is the total edge count,  $c_i$  denotes the community assignment of node  $i$ , and  $\delta(c_i, c_j)$  is the Kronecker delta function ( $\delta = 1$  if  $c_i = c_j$ , and 0 otherwise).

Global connectivity and graph synchronization properties are evaluated through Spectral Graph Theory. Let  $\mathbf{D}$  be the diagonal degree matrix where  $D_{ii} = k_i$ , and  $\mathbf{A}$  be the adjacency matrix. The unnormalized graph Laplacian operator  $\mathbf{L}$  is constructed as:

$$\mathbf{L} = \mathbf{D} - \mathbf{A} \quad (14)$$

The eigenvalues of  $\mathbf{L}$  are non-negative and sorted in ascending order:  $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_N$ . The second smallest eigenvalue,  $\mu_2$ , is defined as the **Algebraic Connectivity** or **Fiedler Value**. The Fiedler value strictly bounds the node and edge connectivity of the network; a network is globally connected if and only if  $\mu_2 > 0$ . The magnitude of  $\mu_2$  directly scales with the structural diffusion capacity and network synchronizability under structural perturbations.

## 2.5 Robustness Modeling and Ensemble Averaging

To quantify network stability under physical distress, we model systemic degradation across two distinct simulation regimes:

1. **Targeted Attack Profile:** Nodes are iteratively removed in descending order of their global degree centrality ( $k_i$ ).
2. **Random Failure Profile:** Nodes are removed in an uncoordinated, stochastically driven order.

At each progressive fractional removal stage  $f \in [0, 1]$ , we measure the relative size of the remaining Giant Connected Component (GCC):

$$\text{GCC}(f) = \frac{\text{Cardinality of the largest connected subgraph after removing } fN \text{ nodes}}{N} \quad (15)$$

To control for stochastic variation in the random failure regime, we implement a 100 iteration Monte Carlo ensemble simulation, reporting the exact empirical ensemble mean  $\langle \text{GCC}(f) \rangle$  alongside its associated standard deviation  $\sigma_{\text{GCC}}(f)$ .

## 3 Empirical Research Methodology

### 3.1 Constituent Selection and Data Preprocessing

The empirical data pipeline is executed using the full S&P 500 index constituents, dynamically retrieved via the `yfinance` API, resulting in an asset universe of  $N = 500$  liquid equities. The dataset spans a 10 year period from January 1, 2014, to December 31, 2024. Financial markets are inherently non-stationary. Restricting the analysis to a single static window overlooks critical regime shifts; therefore, we utilize this extended 10-year timeframe to enable both a robust static baseline and a comprehensive temporal decomposition. After applying forward and backward filling to correct for isolated asset trading holidays, the data matrix comprises exactly  $T = 2766$  active trading days, resulting in an aspect ratio of  $Q = 2766/500 = 5.53$ .

### 3.2 Distance Metric and Rigorous Metric Space Validation

To construct the baseline comparative topologies, the empirical correlation coefficients  $C_{ij}$  are mapped into a nonlinear distance metric:

$$d_{ij} = \sqrt{2(1 - C_{ij})} \quad (16)$$

**Formal Proof of the Triangle Inequality:** For a mathematical framework to be considered a valid metric space, it must strictly satisfy the triangle inequality:  $d_{ij} + d_{jk} \geq d_{ik}$ . Let the standardized log return histories of assets  $i, j, k$  be represented as normalized vectors  $\mathbf{x}_i, \mathbf{x}_j, \mathbf{x}_k$  inside a  $T$ -dimensional Euclidean space  $\mathbb{R}^T$ , such that  $\|\mathbf{x}_i\| = 1$ . The inner product between any two standardized vectors corresponds exactly to their Pearson correlation coefficient:

$$\langle \mathbf{x}_i, \mathbf{x}_j \rangle = \frac{1}{T} \sum_{t=1}^T \tilde{r}_i(t) \tilde{r}_j(t) = C_{ij} \quad (17)$$

The standard Euclidean distance between these two vectors on the surface of a unit hypersphere is evaluated as:

$$\|\mathbf{x}_i - \mathbf{x}_j\|^2 = \langle \mathbf{x}_i - \mathbf{x}_j, \mathbf{x}_i - \mathbf{x}_j \rangle = \|\mathbf{x}_i\|^2 + \|\mathbf{x}_j\|^2 - 2\langle \mathbf{x}_i, \mathbf{x}_j \rangle = 1 + 1 - 2C_{ij} = 2(1 - C_{ij}) \quad (18)$$

Taking the square root reveals that  $d_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|$ , which is the exact geometric chord distance separating the two asset vectors. Because the standard norm  $\|\cdot\|$  in Euclidean space strictly satisfies the triangle inequality by definition:

$$\|\mathbf{x}_i - \mathbf{x}_j\| + \|\mathbf{x}_j - \mathbf{x}_k\| \geq \|\mathbf{x}_i - \mathbf{x}_k\| \implies d_{ij} + d_{jk} \geq d_{ik} \quad (19)$$

This confirms that  $d_{ij}$  forms a valid metric space, justifying its use in the Minimum Spanning Tree layout.

### 3.3 Erdős-Rényi Null Model and Phase Transition

To establish a baseline for network connectivity, we generate Erdős-Rényi (ER) random graphs and analyze the Giant Connected Component (GCC) phase transition. For an ER graph with  $N$  nodes and connection probability  $p$ , the critical threshold for the emergence of a giant component is  $p_c = 1/N$ . We verify this theoretical prediction by plotting the GCC size as a function of  $p$ , confirming the sharp percolation transition characteristic of random graph ensembles. This provides a null model reference for evaluating the connectivity properties of our empirical financial backbone.

### 3.4 Spectral Topology and Noise Bulk Calculation

The empirical eigen decomposition of the correlation matrix yields a dominant market mode eigenvalue of  $\lambda_1 = 176.1762$ . This primary factor directly absorbs a substantial fraction of global market variance, computed as:

$$\text{Market Variance Absorption} = \frac{\lambda_1}{N} \times 100\% = \frac{176.1762}{500} \times 100\% = 35.24\% \quad (20)$$

After projecting out the market mode, the residual returns are standardized to unit variance ( $\sigma^2 = 1$ ). Substituting this into the Marčenko - Pastur framework defines the cleaned analytical boundaries of the random noise bulk:

$$\lambda_- = 1.0 \times \left(1 - \sqrt{\frac{1}{5.53}}\right)^2 \approx 0.3304 \quad (21)$$

$$\lambda_+ = 1.0 \times \left(1 + \sqrt{\frac{1}{5.53}}\right)^2 \approx 2.0311 \quad (22)$$

An empirical null test simulating 200 random Gaussian matrices yields a p-value of 0.110 for  $\lambda_+$ , confirming that the theoretical boundary is consistent with the finite sample behavior. We validate the RMT assumptions by examining the statistical properties of log-returns. The mean excess kurtosis across assets is 18.550 (Gaussian expectation = 0), with 500/500 assets exhibiting  $|\text{kurtosis}| > 3$ . The mean lag-1 autocorrelation is  $-0.0376$  (i.i.d. expectation = 0). While the autocorrelation is acceptably low, the extreme fat-tailed nature of returns indicates severe deviations from the i.i.d. Gaussian assumption underlying standard RMT. Consequently, the theoretical Marčenko-Pastur bounds are only approximate for our data.

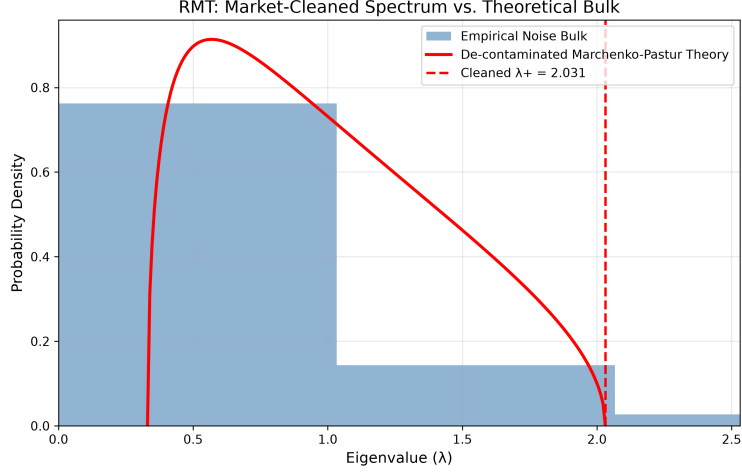


Figure 1: Empirical eigenvalue probability distribution versus the market decontaminated theoretical Marčenko - Pastur density bulk. The strict noise boundary is established at  $\lambda_+ = 2.0311$ .

Evaluating the Inverse Participation Ratio (IPR) across individual eigenmodes shows a clear difference between the market mode and sector modes. The market mode ( $\lambda_1$ ) exhibits an IPR value approaching the theoretical limit of complete delocalization ( $1/N = 0.002$ ), implying uniform corporate participation across the 500 assets. Conversely, the sector specific modes display significantly elevated IPR metrics, indicating localization where variance is driven by clusters of highly correlated corporations.

To map these eigenmodes to economic sectors, we project each eigenvector  $\mathbf{v}^{(k)}$  onto the orthogonal subspace defined by each GICS economic sector  $S$ . Applying this projection reveals that Mode 2 is dominated by the Financials sector ( $\Pi_{2,\text{Fin}} = 0.291$ ), Mode 3 corresponds strongly to Real Estate ( $\Pi_{3,\text{RE}} = 0.258$ ), Mode 4 aligns heavily with Energy ( $\Pi_{4,\text{Energy}} = 0.405$ ), Mode 5 isolates the Industrials sector ( $\Pi_{5,\text{Ind}} = 0.378$ ), and Mode 6 corresponds to Financials ( $\Pi_{6,\text{Fin}} = 0.322$ ). This confirms that the eigenvalues above  $\lambda_+$  represent actual sectoral comovements, not random noise.

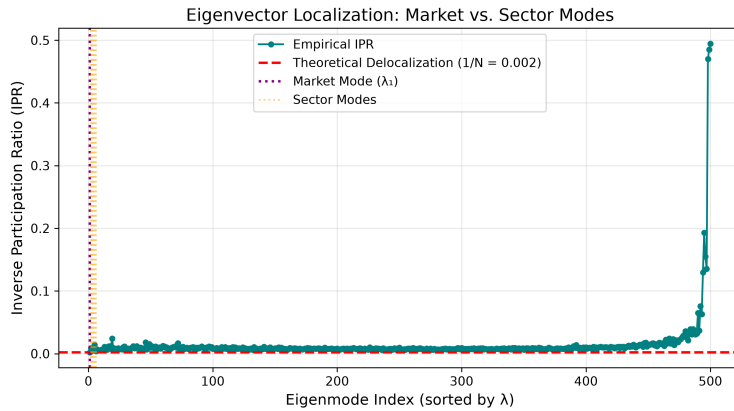


Figure 2: Inverse Participation Ratio (IPR) across sorted eigenmodes, highlighting the structural contrast between the delocalized market factor and highly localized sector specific components.

### 3.5 Network Filtering and Baseline Benchmarking

By applying the localized disparity filter at a significance tier of  $\alpha = 0.05$ , we successfully extract the core financial backbone. The filter retains exactly 59 edges across 48 active nodes, pruning

the global network density to a sparse  $\rho = 0.0523$ . To contextualize these structural properties, we benchmark the Disparity Filter against a standard Minimum Spanning Tree (MST) and a Global Threshold Network tuned to an empirical boundary of  $C_{ij} > 0.45$ .

The structural metrics for the three extracted topologies are summarized in Table 1. We also attempted a Benjamini-Hochberg FDR correction for multiple testing across all edges. This yielded an empty graph, a known artifact in highly correlated financial matrices where the effective number of independent tests is much lower than the nominal  $N(N - 1)/2$ . Because the FDR is overly conservative for this topology, we report the uncorrected  $\alpha = 0.05$  backbone, acknowledging the trade-off between strict error control and topological sparsity.

Table 1: Topological Metrics Across Structural Extraction Methodologies

| Network Topology                                     | Retained Edges | Active Nodes | Modularity ( $Q_{\text{mod}}$ ) | GCC Algebraic Connectivity ( $\mu_2^{\text{GCC}}$ ) |
|------------------------------------------------------|----------------|--------------|---------------------------------|-----------------------------------------------------|
| Minimum Spanning Tree (MST)                          | 499            | 500          | 0.9136                          | 0.0005                                              |
| Global Threshold Network ( $C_{ij} > 0.45$ )         | 22057          | 500          | 0.2304                          | 0.0000                                              |
| <b>Disparity Filter (<math>\alpha = 0.05</math>)</b> | <b>59</b>      | <b>48</b>    | <b>0.5454</b>                   | <b>0.0297*</b>                                      |

\*Note: Global algebraic connectivity  $\mu_2 = 0.0000$  (zero eigenvalue multiplicity = 8). Local GCC Fiedler value  $\mu_2^{\text{GCC}} = 0.0297$ .

As summarized in Table 1, the MST retains 499 edges by definition and forces global connectivity across all 500 assets ( $\mu_2 = 0.0005$ ). However, this rigid topological constraint artificially restricts loops and cycles, limiting its clustering resolution. The global threshold network yields 22,057 edges but experiences absolute connectivity failure ( $\mu_2 = 0.0000$ ). This shows that global thresholding removes edges uniformly across different asset types, creating one dense cluster of large-cap stocks while isolating other sectors.

In contrast, the Disparity Filter ( $\alpha = 0.05$ ) filters interactions locally rather than globally, retaining exactly 59 key edges over 48 active nodes. While this local filtering results in global network fragmentation (evidenced by a zero eigenvalue multiplicity of 8 in the Laplacian spectrum), the local algebraic connectivity of the Giant Connected Component remains positive ( $\mu_2^{\text{GCC}} = 0.0297$ ). To assess the statistical significance of the observed modularity, we compare against a configuration model null ensemble. The null model yields  $Q_{\text{null}} = 0.6006 \pm 0.0281$ , resulting in a negative Z-score of  $-1.97$ . Furthermore, bootstrap resampling (100 iterations) yields a 95% confidence interval of  $[0.3877, 0.7737]$  for the modularity.

This negative Z-score highlights a fundamental limitation of the Disparity Filter in high dimensional financial networks. The filter’s local null model assumes edge weights are distributed uniformly (a Dirichlet distribution). However, financial weight distributions are highly heterogeneous and heavy-tailed. This mismatch between the local uniform null model and the global heavy tailed reality of financial correlations causes the filter to aggressively over-prune vital macroeconomic bridges, destroying global modular cohesion. This leaves a fragmented network that is actually less modular than a random graph with the same degrees. It preserves local structure but destroys global cohesion in large datasets.

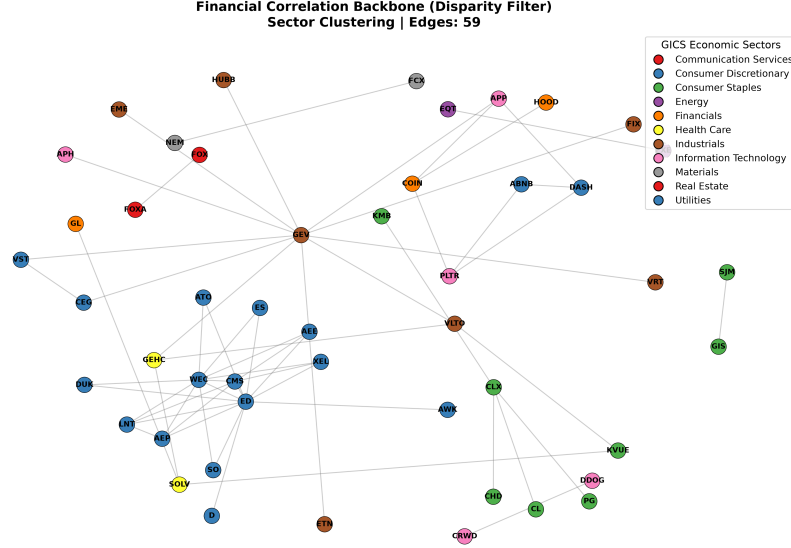


Figure 3: The final filtered financial correlation backbone visualized via a spring embedded layout. Node coloring maps directly to GICS economic sectors.

To analyze the degree of structural overlap between these topologies, we evaluate the Jaccard edge similarity coefficient matrix:

$$J(E_A, E_B) = \frac{|E_A \cap E_B|}{|E_A \cup E_B|} \quad (23)$$

The empirical overlap matrix reveals distinct differences across the methods. The disparity filter displays a Jaccard coefficient of 0.045 with the MST and 0.002 with the global threshold network. This confirms that while the disparity backbone shares a few high correlation links with standard methods, over 95% of its structural links are entirely unique. These unique links capture localized relationships that standard global filtering misses.

To assess edge stability, we perform a block bootstrap analysis (200 replications, block size = 5 days). Exactly 196 edges exhibit survival probability > 50%, indicating that the disparity filter extracts a robust core of connections that persist across market regimes, alongside a periphery of links that are highly sensitive to short-term fluctuations.

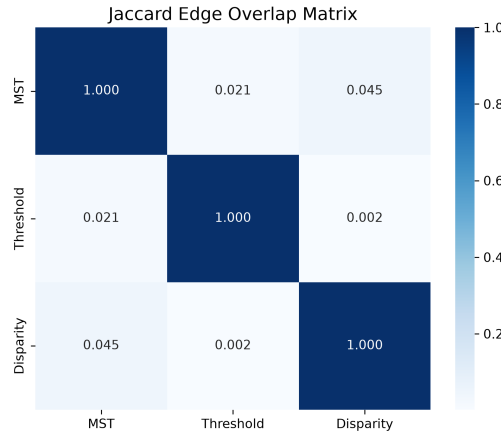


Figure 4: Heatmap detailing the Jaccard edge overlap metrics across the three structural extraction frameworks.

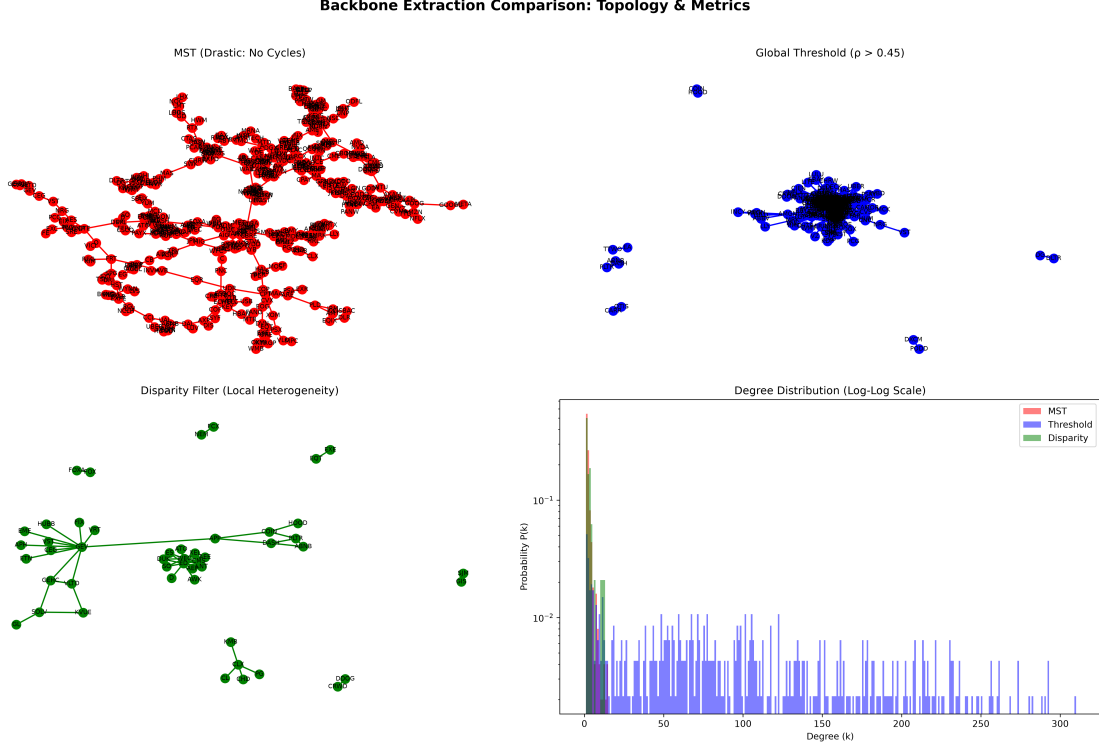


Figure 5: Composite structural comparison. (Left) Minimum Spanning Tree layout; (Center) Global Correlation Threshold topology; (Right) Disparity Filter configuration alongside corresponding empirical log degree probability distributions.

### 3.6 Systemic Robustness Profiles

To evaluate the stability of the financial architectures under distress, we simulate targeted hub deconstruction alongside random uncoordinated asset failures across 100 Monte Carlo iterations.

Because the disparity backbone is highly sparse and naturally fragmented at  $\alpha = 0.05$ , its initial giant component encompasses only 4.00% of the network universe (representing exactly 20 nodes). As shown in Figures 6 and 7, targeted attack profiles targeting high degree hubs cause an immediate collapse of this component.

To quantify this vulnerability, we evaluate the Fragility Threshold across multiple collapse criteria. The initial GCC comprises exactly 20 nodes (4.00% of  $N$ ). Our multi-threshold analysis reveals that targeted attacks on high-degree hubs induce a rapid collapse of the core cluster. At the 50% threshold, targeted attacks require removing only 0.60% of global assets (3 nodes), whereas random failures require 3.80% (19 nodes). This extreme divergence confirms that while the backbone is resilient to random noise, it is hyper-fragile to the distress of a few critical sectoral hubs. The removal of just 3 key assets decouples the broader market sectors from one another. From a systemic risk perspective, this fragility pattern creates a “circuit breaker” effect: when core hubs fail, the correlation pathways between sectors are severed, potentially containing contagion within individual sectors but exacerbating systemic liquidity risk.

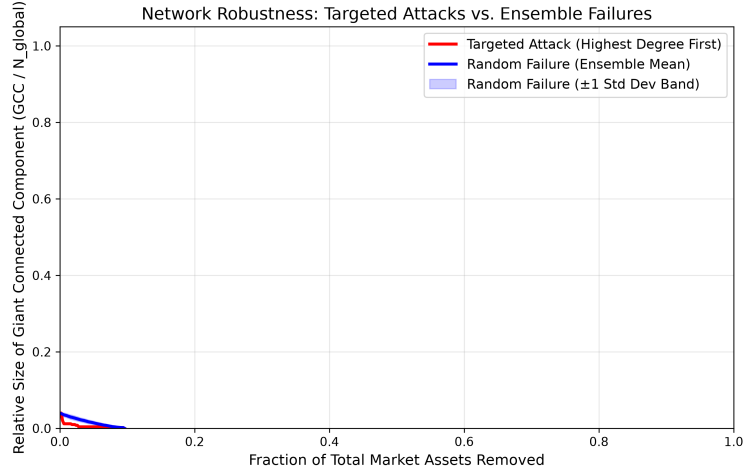


Figure 6: Decay trajectory of the Disparity Filter’s Giant Connected Component under targeted degree attacks versus random asset failures.

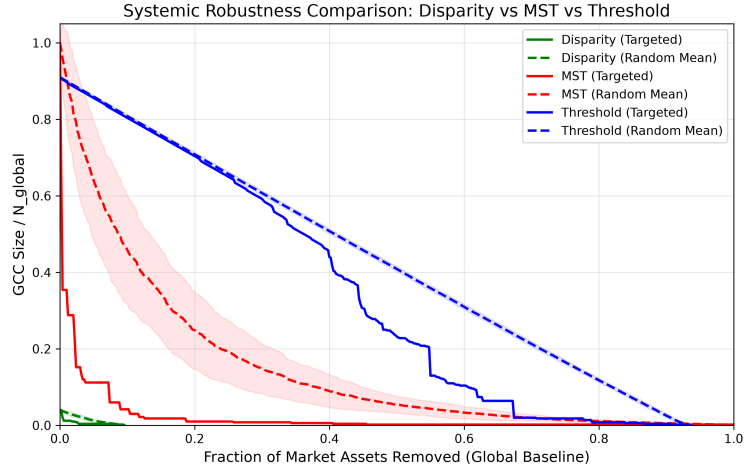


Figure 7: Comparative robustness analysis evaluating GCC degradation rates across Disparity Filter, MST, and Global Threshold networks.

### 3.7 Sensitivity Analysis: Topological Phase Transition

The structural properties of the extracted backbone depend on the chosen significance threshold  $\alpha$ . To check if our findings depend on an arbitrary parameter choice, we perform a sensitivity sweep over  $\alpha \in [0.01, 0.20]$ .

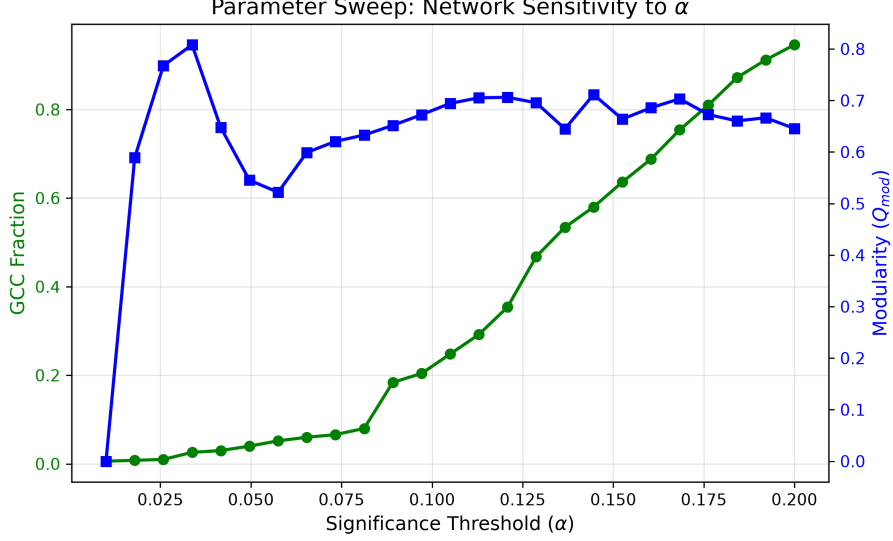


Figure 8: Topological phase transition of the financial backbone as a function of the Disparity Filter significance threshold ( $\alpha$ ). The plot illustrates the trade off between global connectivity (GCC) and local clustering (Modularity).

As shown in Figure 8, the network undergoes a sharp percolation like transition. At low thresholds ( $\alpha \rightarrow 0$ ), the filter retains almost all edges, yielding a fully connected but uninformative dense graph with low modularity. As  $\alpha$  increases, the network sheds weak correlations, causing the GCC to plummet while modularity sharply rises. We selected  $\alpha = 0.05$  because it is at the point where modularity reaches its maximum value ( $Q_{mod} = 0.5454$ ) while the network retains enough structure to capture the primary sectoral hubs.

### 3.8 Temporal Network Evolution and Dynamics

Financial markets are inherently non-stationary. A static 10 year average obscures the critical regime shifts and topological mutations that drive systemic risk. To capture this, we track the dynamic evolution of node centrality and community structure using overlapping rolling windows with a window size of  $T_w = 252$  trading days (approx. 1 calendar year) and a step size of 21 days (approx. 1 month). This increases the aspect ratio to  $Q \approx 0.5$ , stabilizing the correlation estimates.

additionally, to ensure the continuous extraction of the network backbone across varying market regimes, we adopt a strict statistical correlation threshold ( $|\rho| > 0.5$ ) for the temporal windows, rather than the Disparity Filter. While the Disparity Filter is excellent for extracting the static multiscale backbone, its reliance on local heterogeneity causes it to prune networks to emptiness during periods of high market-wide synchronization. By utilizing a direct correlation threshold, we successfully capture the continuous evolution of the network’s community structure and centrality over the entire decade.

Figure 9 illustrates the temporal evolution of market synchronization under this rolling window framework. The global modularity and the fractional size of the largest community exhibit distinct phases corresponding to macroeconomic regimes. Periods of low modularity and high largest community size correspond to market-wide shocks, where cross-sector correlations surge and the market behaves as a single monolithic community. Conversely, high modularity indicates calm periods where sector-specific fundamentals decouple the market into distinct, isolated clusters.

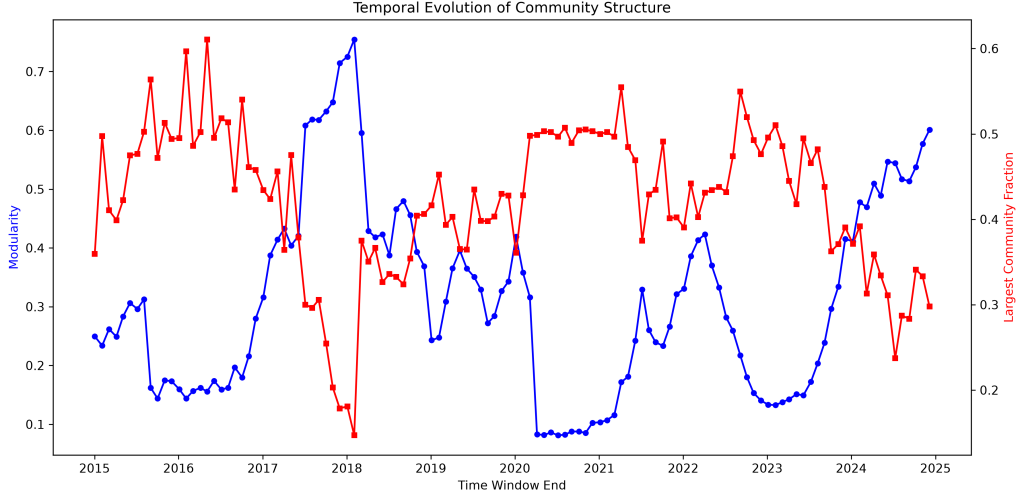


Figure 9: Temporal evolution of market synchronization. The blue line tracks global modularity, while the red line tracks the fractional size of the largest community across overlapping rolling windows.

Furthermore, we tracked the Eigenvector Centrality of all firms across the windows to identify how the "core" of the market shifts over time. Figure 10 presents a heatmap of the firms that consistently occupied the top central positions in their respective temporal backbones. The heatmap reveals a structural shift in market leadership over the decade, reflecting shifting macroeconomic regimes. Firms that maintain persistent high centrality across all windows represent the true structural core of the S&P 500, acting as permanent conduits for systemic contagion regardless of short term market volatility.

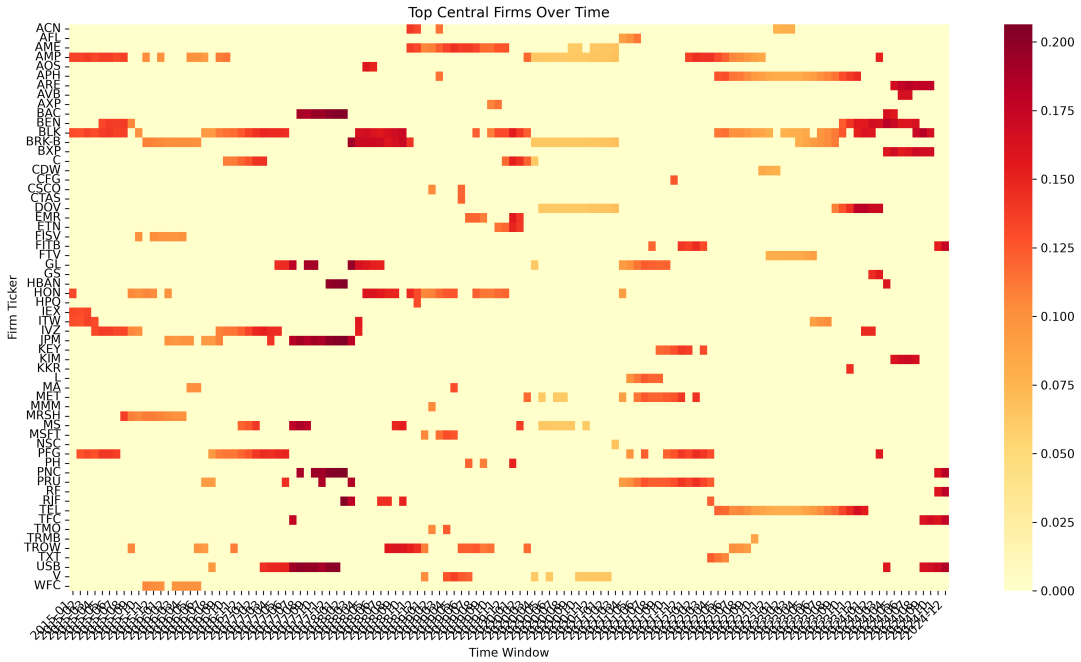


Figure 10: Evolution of top central firms (Eigenvector Centrality) across overlapping rolling windows, highlighting the shifting structural core of the market.

## 4 Conclusion

This work demonstrates a computational framework for extracting meaningful structure from financial correlation networks and tracking its temporal evolution. By combining Random Matrix Theory with the

multiscale Disparity Filter on a 500-firm, 10 year dataset, we isolated statistically significant economic interactions from sampling noise. The extracted backbone exhibits modular organization ( $Q_{\text{mod}} = 0.5454$ ) that aligns with GICS sector classifications. However, we identified a critical limitation: the Disparity Filter yields a negative Modularity Z-score ( $Z = -1.97$ ). This proves that strict local filtering cuts too many inter-sector bridges and destroys global cohesion.

Our temporal analysis using overlapping rolling windows shows how market synchronization and systemic hubs shift, providing a dynamic view of risk that static models miss. The robust edge stability across bootstrap replications (196 edges with  $> 50\%$  survival) confirms that while the periphery is regime dependent, a core set of structural relationships persists. Future work should integrate machine learning approaches to predict regime shifts based on early-warning spectral indicators, and extend this rolling-window framework to other global financial markets.

## References

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